

# **equation**

Algebra 1 • 2.1

# **solution**

Algebra 1 • 2.1

# **solve**

Algebra 1 • 2.1

# **translate**

Algebra 1 • 2.1

# **inverse operations**

Algebra 1 • 2.2

# **isolate the variable**

Algebra 1 • 2.2

# **equivalent equations**

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# **Addition Property of Equality**

Algebra 1 • 2.2

A value of the variable that makes an equation true.

$x = 6$  is the solution of  $3x - 7 = 11$ .

A mathematical sentence stating that two expressions are equal.

$$3x - 7 = 11$$

To turn a verbal sentence into an equation.

“Five less than twice a number is 9”  $\rightarrow 2n - 5 = 9$

To find every value of the variable that makes the equation true.

To get the variable alone on one side of the equation using inverse operations.

Operations that undo each other.

$+$  and  $-$ ;  $\times$  and  $\div$

You may add the same number to both sides:  
if  $a = b$ , then  $a + c = b + c$ .

Equations that have the same solution(s).

$$2x = 10 \text{ and } x = 5$$

# **Subtraction Property of Equality**

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# **Multiplication Property of Equality**

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# **Division Property of Equality**

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# **multi-step equation**

Algebra 1 • 2.3

# **consecutive integers**

Algebra 1 • 2.3

# **identity**

Algebra 1 • 2.4

# **no solution**

Algebra 1 • 2.4

# **absolute value equation**

Algebra 1 • 2.5

You may multiply both sides by the same number: if  $a = b$ , then  $ac = bc$ .

You may subtract the same number from both sides: if  $a = b$ , then  $a - c = b - c$ .

An equation that takes more than one operation to solve.

Undo in REVERSE order of operations:  $4(x - 3) + 2 = 18$

You may divide both sides by the same nonzero number: if  $a = b$  and  $c \neq 0$ , then  $a/c = b/c$ .

An equation that is true for EVERY value of the variable. Solving it gives a statement that is always true.

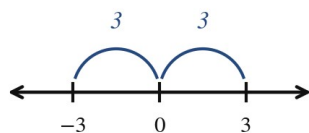
$$2(x + 3) = 2x + 6 \rightarrow 6 = 6$$

Integers that follow one another in order.

$n, n + 1, n + 2$  (consecutive EVEN or ODD:  $n, n + 2, n + 4$ )

An equation containing an absolute value expression. It usually has TWO solutions.

$$\begin{aligned} |x - 3| = 5 &\rightarrow x - 3 = 5 \\ \text{or } x - 3 &= -5 \rightarrow x = 8 \\ \text{or } x &= -2 \end{aligned}$$



An equation that is true for NO value of the variable. Solving it gives a false statement.

$$3x + 7 = 3x - 2 \rightarrow 7 = -2$$

# empty set ( $\emptyset$ )

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# ratio

Algebra 1 • 2.6

# rate

Algebra 1 • 2.6

# unit rate

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# proportion

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# cross products

Algebra 1 • 2.6

# percent

Algebra 1 • 2.6

# literal equation

Algebra 1 • 2.7

A comparison of two quantities by division.  
3 to 5, 3:5, or  $3/5$

The set with no elements. An equation like  $|x| = -4$  has no solution, so its solution set is  $\emptyset$ .

A rate with a denominator of 1.  
60 miles per hour

A ratio that compares quantities with different units.  
120 miles in 2 hours

In the proportion  $a/b = c/d$ , the cross products are  $ad$  and  $bc$ . If the ratios are equal, the cross products are equal:  $ad = bc$ .

An equation stating that two ratios are equal.  
 $x/4 = 9/12$

An equation with two or more variables — often a formula. Solve it for one variable in terms of the others.  
Solve  $P = 2l + 2w$  for  $w$ :  $w = (P - 2l)/2$

A ratio that compares a number to 100.  
 $35\% = 35/100 = 0.35$

# formula

Algebra 1 • 2.7

# dimensional analysis

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# error analysis

Algebra 1 • 2.3

Carrying units through a calculation and canceling them like factors to convert or check an answer.

$$60 \text{ mi/hr} \times 1 \text{ hr}/60 \text{ min} = 1 \text{ mi/min}$$

An equation that states a rule for the relationship between certain quantities.

$$A = lw, \quad d = rt, \quad F = (9/5)C + 32$$

Finding the mistake in a worked solution and explaining how to fix it.